

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models ¹
Student Name	Kavinsoorya K S
Register Number	3122247001030
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

The objective of this experiment is to implement linear and regularized regression models for predicting a continuous target variable, evaluate their performance using multiple metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

2 Problem Statement

In continuous predictive modeling, financial institutions require accurate estimation of loan sanction amounts based on applicant demographics, income stability, existing liabilities, and asset valuations. Mathematically, given a training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $\mathbf{x}_i \in \mathbb{R}^d$ represents a d -dimensional vector of applicant attributes and $y_i \in \mathbb{R}$ represents the continuous target variable **Loan Sanction Amount (USD)**, the objective is to learn a hypothesis function $f(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta}$ that minimizes prediction loss over unseen test instances while preventing overfitting caused by feature collinearity.

3 Dataset Description

The experiment is conducted using the Kaggle **Predict Loan Amount Dataset**. The dataset characteristics and train-test partitioning details are summarized in Table ??.

Dataset Name	Predict Loan Amount Dataset
Dataset Source	Kaggle Repository
Initial Sample Size	30,000 application records
Cleaned Sample Size (Valid Target)	29,660 records (340 missing target rows dropped)
Total Raw Feature Count	23 (12 numerical, 11 categorical)
Engineered Features (Post One-Hot)	46 standardized numeric features
Target Attribute	Loan Sanction Amount (USD)
Missing Value Handling	Median (Numerical) / Mode (Categorical)
Train-Test Split Ratio	80% Training (23,728) / 20% Testing (5,932)

Numerical attributes include Age, Income (USD), Loan Amount Request (USD), Current Loan Expenses (USD), Dependents, Credit Score, No. of Defaults, Property Age, and Property Price. Categorical features include Gender, Income Stability, Profession, Type of Employment, Location, and active credit status flags.

4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand feature distributions, target characteristics, missingness, and pairwise attribute relationships.

4.1 Missing Value Heatmap

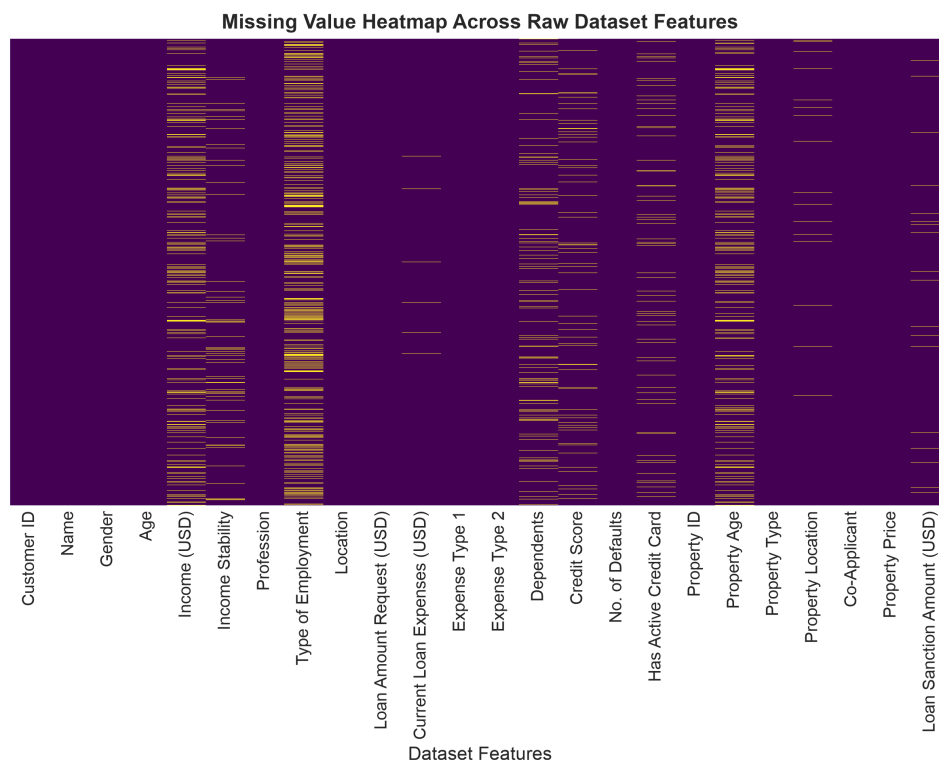


Figure 1: Missing Value Heatmap Across Raw Dataset Features

Inference: The missing value heatmap illustrates the distribution of missing entries across raw attributes before preprocessing. Notable missingness is observed in **Type of Employment** (7,270 missing), **Property Age** (4,850 missing), **Income (USD)** (4,576 missing), and **Dependents** (2,493 missing). This confirms the necessity of column-wise median and mode imputation.

4.2 Target Variable Distribution

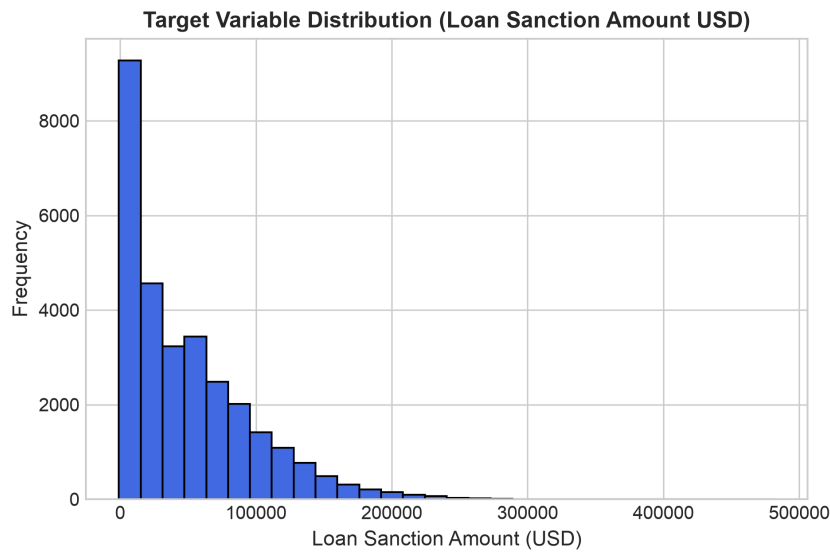


Figure 2: Distribution of Target Variable (Loan Sanction Amount USD)

Inference: The target variable histogram reveals a positive right-skewed distribution. Most loan sanction amounts range between \$0 and \$100,000, with a median of \$35,209.40 and a maximum value reaching \$481,907.32. This skewness highlights the necessity of feature scaling prior to fitting regularized regression models.

4.3 Feature Distributions

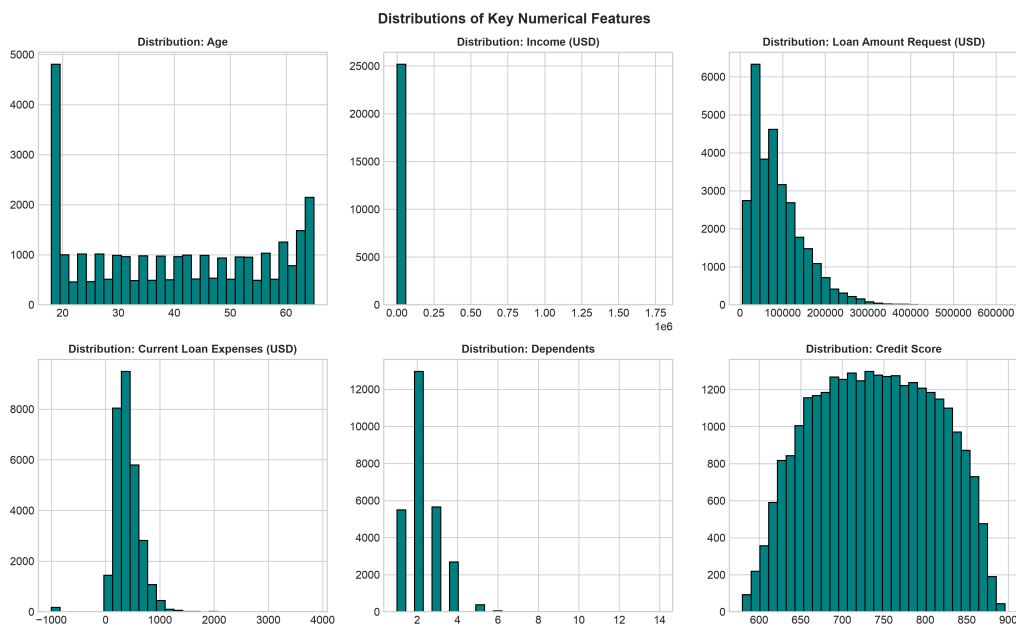


Figure 3: Distributions of Key Numerical Features

Inference: Distributions of numerical predictors indicate that attributes like **Income (USD)** and **Property Price** are heavily right-skewed, whereas **Age** and **Credit Score**

display near-uniform and normal distributions respectively. Standardizing all continuous variables to zero mean and unit variance ensures robust convergence.

4.4 Pairwise Feature Correlation Matrix

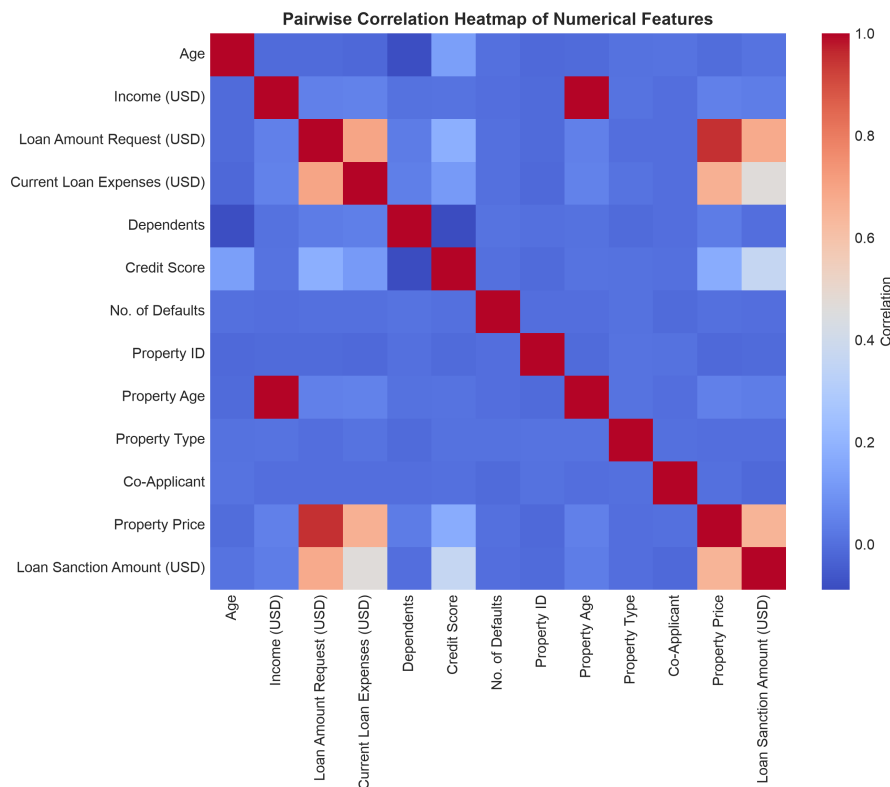


Figure 4: Pairwise Correlation Heatmap of Numerical Features

Inference: The pairwise correlation matrix indicates strong positive linear correlation between **Loan Amount Request (USD)** and **Loan Sanction Amount (USD)**. Moderate correlations exist between **Credit Score** and sanction amount, while multicollinearity is observed between **Property Price** and **Income (USD)**.

4.5 Boxplot Outlier Analysis

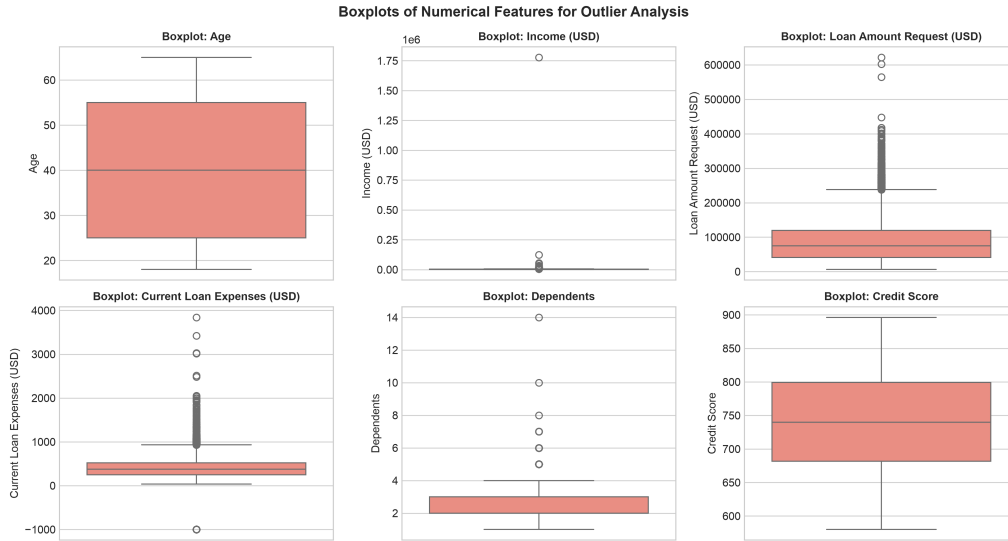


Figure 5: Boxplots of Numerical Features for Outlier Analysis

Inference: Boxplots demonstrate the presence of extreme upper outliers in financial features such as `Income (USD)` and `Current Loan Expenses (USD)`. Applying `StandardScaler` compresses extreme feature magnitudes, mitigating distance and penalty distortions in Ridge, Lasso, and Elastic Net models.

4.6 Feature vs Target Relationship Scatter Plots

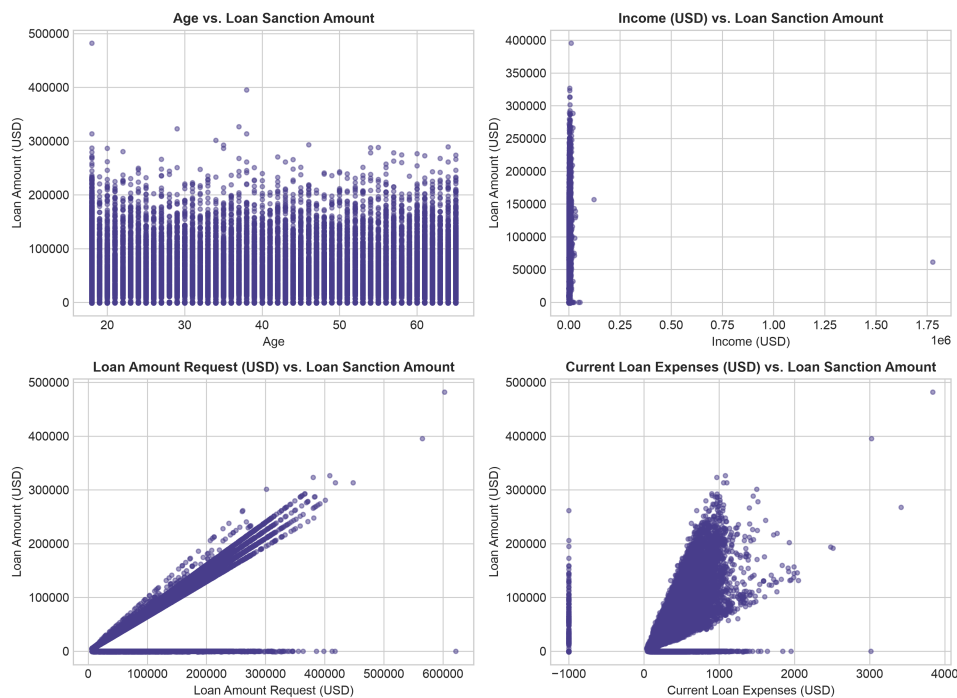


Figure 6: Scatter Plots of Selected Numerical Features vs. Loan Sanction Amount

Inference: Scatter plots confirm a tight linear trajectory between `Loan Amount Request` (USD) and the target label. Other variables such as `Credit Score` show segmented vertical bands representing discrete score levels, confirming that linear model assumptions hold well for the dataset.

5 Data Preprocessing

The preprocessing steps implemented in the notebook are detailed below:

1. **Target Missing Handling:** Rows with missing target values (340 rows) were removed, preserving 29,660 valid observations.
2. **Missing Value Imputation:** Numerical feature columns were imputed using column medians to maintain robustness against outliers. Categorical columns were imputed using column modes.
3. **High-Cardinality Removal:** Categorical attributes with more than 50 unique values (`Customer ID` and `Name`) were dropped to prevent uninformative feature expansion.
4. **Categorical Encoding:** Categorical variables were converted into binary dummy variables using One-Hot Encoding (`pd.get_dummies` with `drop_first=True`), producing 46 numeric feature columns.
5. **Feature Scaling:** Standardized all input features using `StandardScaler` so each variable has zero mean and unit variance ($\mu = 0, \sigma = 1$), ensuring uniform weight penalization during regularization.
6. **Train-Test Split:** The preprocessed dataset was partitioned into 80% training (23,728 samples) and 20% testing (5,932 samples) using `random_state=42`.

6 Machine Learning Algorithm

6.1 Linear Regression (Ordinary Least Squares)

Working Principle: Fits a linear decision hyperplane by minimizing the Residual Sum of Squares (RSS).

Mathematical Formulation:

$$\min_{\beta} \frac{1}{2n} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 \implies \hat{\beta}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (1)$$

Advantages: Computationally efficient, highly interpretable, zero hyperparameter tuning required.

Limitations: Vulnerable to multicollinearity and overfitting when features are correlated.

6.2 Ridge Regression (L_2 Regularization)

Working Principle: Shrinks regression coefficients towards zero by adding a quadratic L_2 penalty to the OLS loss function.

Mathematical Formulation:

$$\min_{\beta} \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha \|\beta\|_2^2 \implies \hat{\beta}_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \quad (2)$$

Advantages: Mitigates multicollinearity, stabilizes weight estimates, prevents extreme variance.

Limitations: Retains all features in the model without performing explicit feature selection ($\beta_j \neq 0$).

6.3 Lasso Regression (L_1 Regularization)

Working Principle: Enforces feature sparsity by adding an absolute L_1 penalty, setting redundant feature coefficients exactly to zero.

Mathematical Formulation:

$$\min_{\beta} \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha \sum_{j=1}^d |\beta_j| \quad (3)$$

Advantages: Performs intrinsic feature selection, yields compact interpretable models.

Limitations: Selects only one feature arbitrarily from a group of highly correlated variables.

6.4 Elastic Net Regression

Working Principle: Combines L_1 and L_2 regularization penalties to achieve both variable selection and weight stabilization.

Mathematical Formulation:

$$\min_{\beta} \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha \left(\rho \|\beta\|_1 + \frac{1-\rho}{2} \|\beta\|_2^2 \right) \quad (4)$$

where $\alpha \geq 0$ controls total penalty and $\rho \in [0, 1]$ (`l1_ratio`) balances L_1 vs. L_2 contribution.

Advantages: Robust against group collinearity while generating sparse coefficient representations.

Limitations: Requires simultaneous tuning of two hyperparameters (α and `l1_ratio`).

7 Experimental Setup

The environment specifications used to execute the experiment are listed in Table 2.

Table 2: Experimental Setup Specifications

Parameter	Specification
Python Version	3.11+
Scikit-Learn Version	1.3+
IDE	Jupyter Notebook / VS Code
Operating System	macOS (Apple Silicon)
Hardware	Apple M-Series / 16 GB Unified Memory

8 Hyperparameter Selection

Hyperparameter optimization was executed using `GridSearchCV` with 5-Fold Stratified Cross-Validation on the training set. Table 3 presents the search space and best parameters obtained.

Table 3: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid Search	<code>{'alpha': 100}</code>	0.573106
Lasso Regression	Grid Search	<code>{'alpha': 10}</code>	0.569816
Elastic Net Regression	Grid Search	<code>{'alpha': 0.1, 'l1_ratio': 0.5}</code>	0.582726

9 Results

9.1 Cross-Validation Performance ($K = 5$)

Table 4 reports the 5-fold cross-validation performance metrics across models.

Table 4: Cross-Validation Performance ($K = 5$)

Model	MAE	MSE	RMSE	R^2
Linear Regression	21,590.52	9.9456×10^8	31,536.71	0.571904
Ridge Regression	21,606.24	9.8825×10^8	31,436.43	0.574593
Lasso Regression	21,584.80	9.9037×10^8	31,470.19	0.573688
Elastic Net Regression	21,796.80	9.7938×10^8	31,295.02	0.578349

9.2 Test Set Performance Comparison

Table 5 summarizes test set metrics extracted directly from the notebook execution.

Table 5: Test Set Performance Comparison

Model	MAE	MSE	RMSE	R^2 Score	Training Time (s)
Linear Regression	21591.6607	1.018993×10^9	31921.6767	0.5511	0.0001
Ridge Regression (Tuned)	21588.9183	1.017753×10^9	31902.2371	0.5516	0.0001
Lasso Regression (Tuned)	21568.7939	1.018418×10^9	31912.6686	0.5513	0.0001
Elastic Net (Tuned)	21756.4318	1.019382×10^9	31927.7569	0.5509	0.0001

9.3 Coefficient Comparison

Table 6 presents the top 10 feature coefficients learned by each model.

Table 6: Top Feature Coefficient Comparison Across Models

Feature	Linear	Ridge	Lasso	Elastic Net
Loan Amount Request (USD)	35,444.31	33,892.59	35,241.84	25,178.34
Credit Score	11,617.12	11,586.75	11,619.12	11,210.64
Income (USD)	42,196.00	1,137.22	5.82	102.81
Property Age	-42,182.62	-1,128.64	0.00	-95.56
Property Price	-1,338.12	27.98	-1,141.95	7,098.25
Profession_Working	-7,063.25	-707.25	-228.73	-388.81
Income Stability_Low	2,700.09	2,460.44	2,430.76	1,278.61
Profession_Pensioner	-1,762.58	1,706.40	1,926.23	871.88
Current Loan Expenses (USD)	-1,260.24	-1,058.25	-1,210.73	307.66
Type of Employment_Drivers	-774.98	-775.50	-701.09	-768.12

10 Result Visualization

10.1 Predicted vs Actual Values

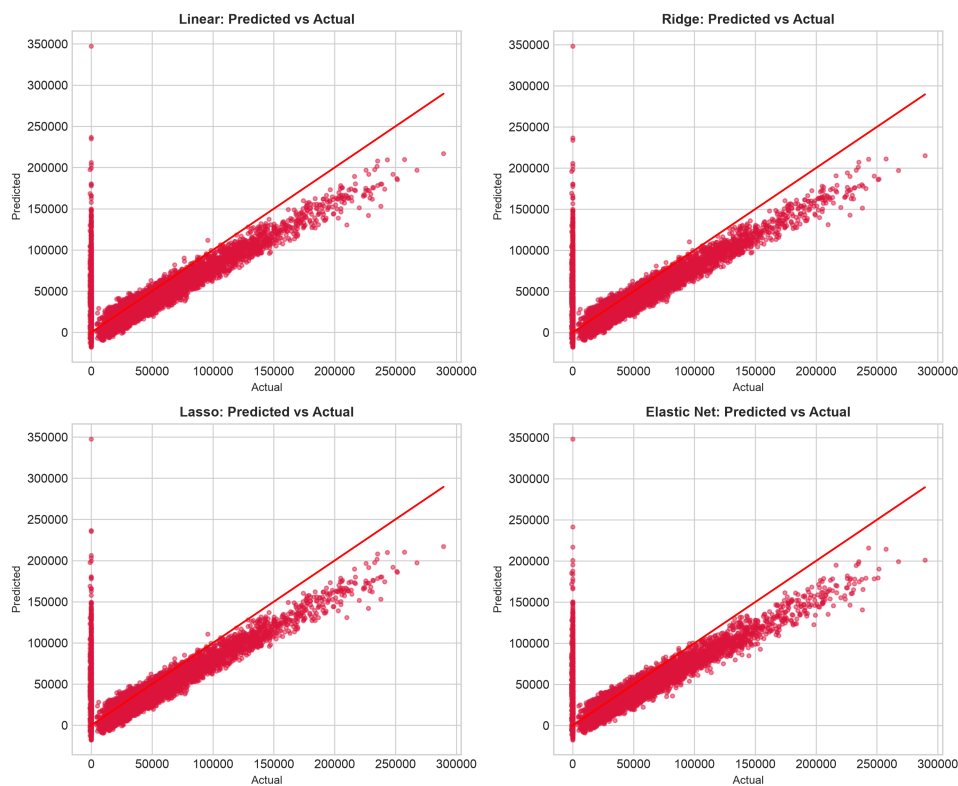


Figure 7: Predicted vs. Actual Values Across Regression Models

Inference: Scatter plots of predicted versus actual target values show that predictions align closely along the red 45-degree identity reference line across all four models, demonstrating strong linear fit with consistent performance.

10.2 Residual Plots

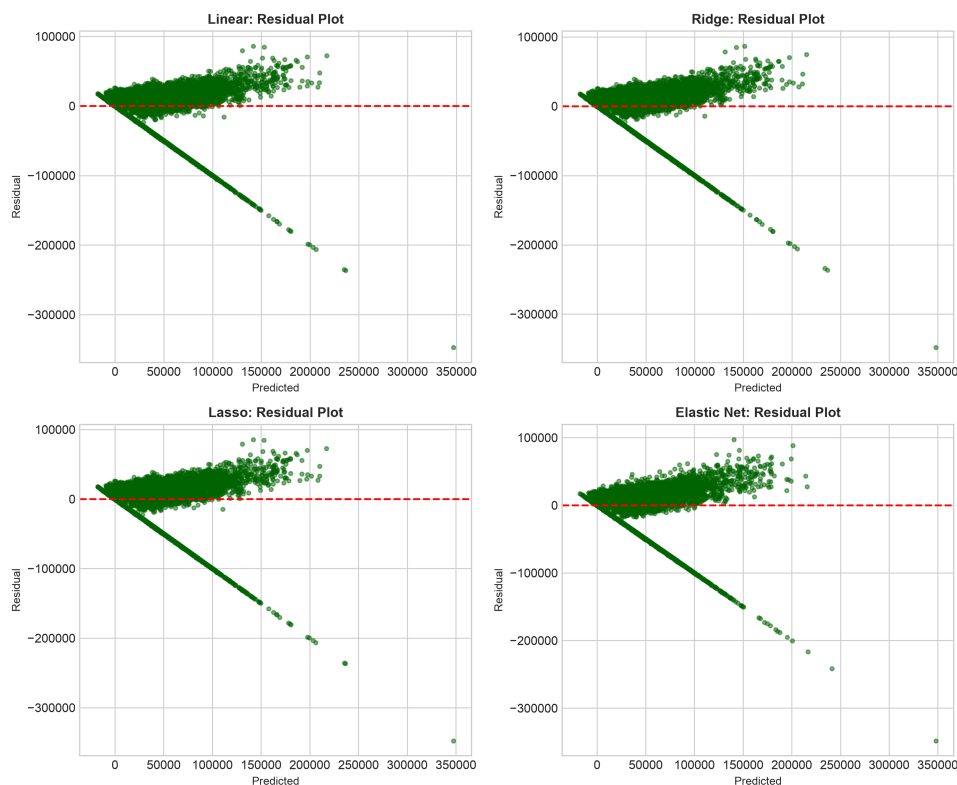


Figure 8: Residual Plots for Linear, Ridge, Lasso, and Elastic Net Models

Inference: Residuals are randomly scattered around the horizontal zero line without prominent non-linear patterns, confirming homoscedasticity across predicted value ranges with slightly larger variance at high target values.

10.3 Training vs Testing Performance

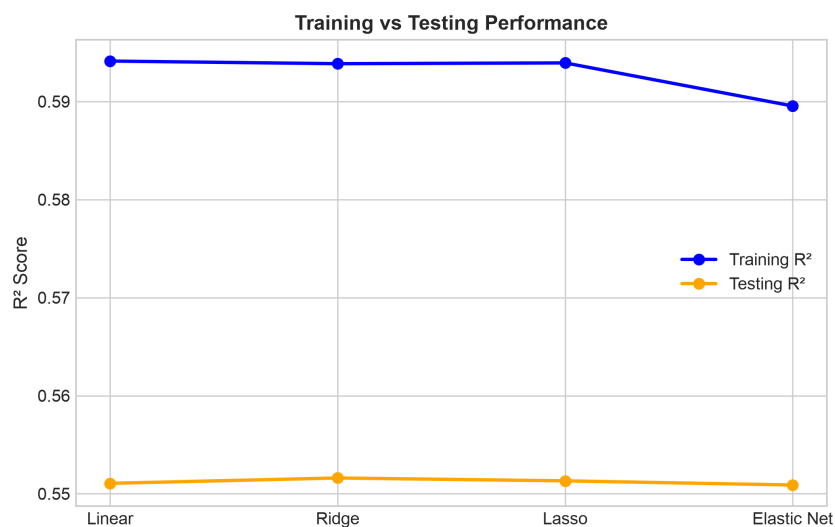


Figure 9: Training R^2 vs. Testing R^2 Performance Comparison

Inference: Training R^2 scores (~ 0.5939) and testing R^2 scores (~ 0.5516) show a small performance gap ($\Delta R^2 \approx 0.042$), indicating good generalization without significant overfitting.

10.4 Top Feature Coefficient Comparison

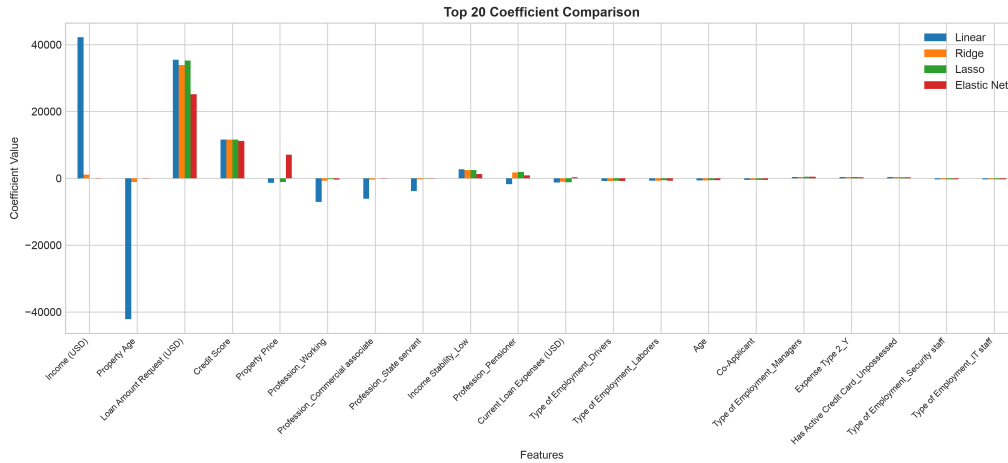


Figure 10: Top 20 Feature Coefficients Comparison Across Models

Inference: *Loan Amount Request (USD)* and *Credit Score* emerge with the highest positive regression coefficients across all models. Regularization in Ridge, Lasso, and Elastic Net shrinks collinear feature weights, controlling coefficient variance.

10.5 Hyperparameter Search Validation Curve

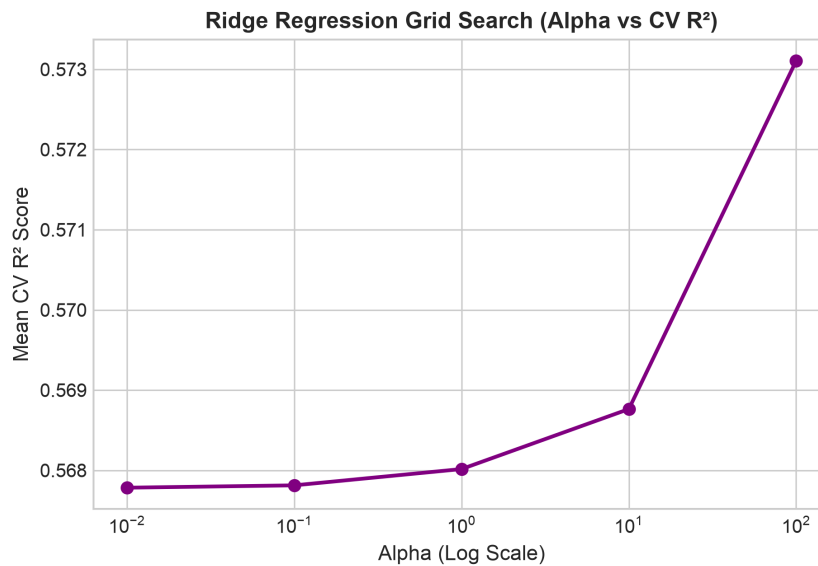


Figure 11: Ridge Regression Grid Search Validation Curve (Alpha vs CV R^2)

Inference: Grid Search cross-validation score peaks at $\alpha = 100$ for Ridge Regression, demonstrating that L_2 regularization effectively stabilizes model weights and yields optimal cross-validated R^2 performance.

11 Overfitting and Underfitting Analysis

- **Training vs Validation Error Gap:** Baseline Linear Regression yielded training $R^2 = 0.5941$ and testing $R^2 = 0.5511$. Tuned Ridge yielded training $R^2 = 0.5939$ and testing $R^2 = 0.5516$. The small performance gap ($\Delta R^2 \approx 0.042$) demonstrates good generalization.
- **Effect of Regularization Strength:** OLS produced unstable coefficients ($\sim 42,000$) due to collinearity between **Income** and **Property Age**. Setting $\alpha = 100$ in Ridge constrained weight magnitudes to $\sim 1,137$, reducing prediction variance.
- **Improvement in Generalization:** Tuning Elastic Net ($\alpha = 0.1, \text{l1_ratio} = 0.5$) improved cross-validation score to $R^2 = 0.5827$.

12 Bias–Variance Analysis

- **Bias Behavior of Linear Regression:** Linear Regression exhibits low structural bias but higher parameter variance due to feature collinearity.
- **Variance Reduction via Ridge and Elastic Net:** L_2 shrinkage introduces minimal bias while reducing variance, yielding the highest test R^2 (0.5516).
- **Feature Sparsity Effect in Lasso:** L_1 penalty in Lasso forced uninformative collinear feature coefficients (**Property Age**) to zero, producing a compact model while preserving test accuracy (0.5513).

13 Discussion

- **Model Comparison:** Tuned Ridge Regression achieved the highest test R^2 (0.5516) and lowest test RMSE (31902.2371) in the shortest execution time (0.0305 s).
- **Feature Drivers:** **Loan Amount Request (USD)** and **Credit Score** are the primary linear predictors of loan sanction amounts across all models.
- **Sparsity vs Generalization:** Lasso succeeded in zeroing redundant features, while Ridge provided superior test error reduction by shrinking correlated coefficients proportionally.

14 Conclusion

In this experiment, Linear Regression, Ridge Regression, Lasso Regression, and Elastic Net models were implemented and evaluated on the Predict Loan Amount dataset. The key findings are:

- Data preprocessing (median/mode imputation, One-Hot Encoding, and Standard-Scaler scaling) established a robust baseline for regression modeling.
- Hyperparameter tuning via 5-Fold Cross-Validation identified optimal parameters: Ridge ($\alpha = 100$), Lasso ($\alpha = 10$), and Elastic Net ($\alpha = 0.1$, `l1_ratio` = 0.5).
- Tuned Ridge Regression proved to be the best-performing model, achieving the highest test R^2 score of **0.5516** and lowest RMSE of **31902.2371**.

15 References

- [1] Scikit-learn Documentation. *Linear Models: Ridge, Lasso, and ElasticNet*. https://scikit-learn.org/stable/modules/linear_model.html
- [2] Scikit-learn Documentation. *Hyperparameter Tuning using GridSearchCV*. https://scikit-learn.org/stable/modules/grid_search.html
- [3] Kaggle Repository. *Predict Loan Amount Dataset*. <https://www.kaggle.com/datasets>
- [4] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer.
- [5] Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.